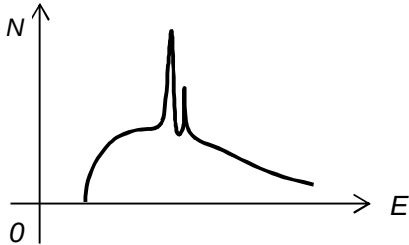
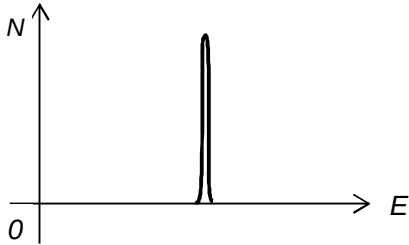


|    |                                                                                                                                              |                                                                                   |   |                                                                                    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---|------------------------------------------------------------------------------------|
| 29 | Which one of the following best represents the number $N$ of alpha-particles emitted from a given radioactive source possessing energy $E$ ? |                                                                                   |   |                                                                                    |
|    | A                                                                                                                                            |  | B |  |
|    | C                                                                                                                                            |  | D |  |

**Answer: D**

In alpha-decay, there are only 2 products: parent nucleus  $\rightarrow$  daughter nucleus + alpha-particle (only one particular reaction, and no other forms of energy).

Hence to ensure conservation of momentum and conservation of energy, each product's speed and kinetic energy would be of fixed values.